ML Basics

The learning process

A basic learning approach can be divided into the following pipeline:

- Data acquisition
- Data preprocessing
- Dimensionality reduction
- Learning
- Model testing

Data Acquisition

- Involves **collecting** relevant **data for the problem**
- Data collection and annotation are **key steps** → **quality and relevance** are **crucial**
- **DEPENDS ON** the application domain

Data Preprocessing

- **Collected data** needs to be **cleaned and prepared** for analysis
  - Handling missing values, dealing with outliers, transforming into a suitable format
- **DEPENDS ON** the application domain

Dimensionality reduction

- Some techniques can be applied to data to r**educe the number of features they represent**
  - In order to **avoid noise and** computational **complexity**
  - **Preserving** the most **important information**
- **DEPENDS ON** the application domains

Learning

- A model is trained on preprocessed data to learn patterns and relationships
  - The **core phase of the ML pipeline**
- The **choice of the model depends on** both the **problem** and the **data**
- **DOESN'T depend on** the application domain

Model Testing

- The **trained model needs to be evaluated**
  - Using a **separate dataset** → the **test set**
    - Not used during the training phase
  - To assess performance and generalization capabilities

Iteration and refinement

It's important to **iterate and refine each step** of the pipeline

- To i**mprove performance and effectiveness** in real-world applications

Data

Training data are built on examples

An **example** is a **set of features**:

- **How the algorithm represents data**
- They are the questions we can ask about the examples
- Are usually **represented with vectors**

Mapping data to features implies a **loss of information**:

- **Features** are usually **derived or selected from original data**
- Only the ones relevant for a particular task or analysis are kept
- → Via **simplification** or **abstraction**

We distinguish:

- **Training example** → an example **used** during the **training phase**
- **Test example** → an example **used** during the **testing phase**

Types of learning

We distinguish 3 main forms of ML techniques:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

SUPERVISED LEARNING

- **Training examples are labeled** (= the user explicitly sets their features)
- The model's aim is to **predict a label for** a given **test example**

Classification

Categorize input data **into** predefined **classes or categories**

Regression

Predict continuous numerical values (or vectors of in case it's multidimensional) based on input

Ranking

Order a set of items based on relevance or preference

UNSUPERVISED LEARNING

- **Training examples aren't labeled**
- The model's aim is to **predict a label for** a given **test example**

Clustering

Group similar data points **into clusters** based on features or inherited characteristics

Dimensionality Reduction

Reduce the **number of features** while **preserving** the most **important information**

Density Estimation

Find a **probability distribution** that **fits** the **data**

REINFORCEMENT LEARNING

- An agent **learns from the environment**
  - **Interacting** with it, executing a set of predefined actions
  - **Receiving rewards** for performing specific actions or reaching specific events

Other learning variations

SEMI-SUPERVISED LEARNING

- Training examples can be both labeled and unlabeled

ACTIVE LEARNING

- Training examples are labeled
- The model analyses unlabeled data and selects which should get a label
  - The oracle (can also be a human user) assigns the labels to the selected data
  - The model gets trained again with the new examples

Online and Offline learning

2 approaches exists **in the learning process**:

- **Offline** (Batch) **Learning**
  - **The model learns** the **whole dataset** in one go → updated after processing all data
  - **Once trained**, the model **doesn't change**
- **Online Learning**
  - **The model learns** in an **incremental** way → updated after every new data point is analysed
  - It **adapts to the changes** in the data distribution

Generalization

Learning is about generalizing from the training data

- Is the **ability** of a trained model **to perform well on new and unseen data** → make predictions

To achieve generalization, the **training** and the **test sets need to be similar**

- Testing data are expected **to have similar features** to the training ones

We talk about a probabilistic model of learning → **Data generating distribution**

- Probability **distribution over example and label pairs**
- **Both** the training and the test **sets** are generated **based on it**
- It is **unknown**

Lazy or eager learner

We distinguish ML algorithms in 2 categories:

LAZY LEARNING: **stores training data** and **operates** when it's **given** a **test example**

EAGER LEARNING: **given** a **training set** immediately **constructs** a **model**

Probability distribution

Is a **mathematical function**

- Represents and quantifies uncertainty
- Tells **how probable it is for different events to occur**

Discrete Probability Distribution

- The probability has specific values
- We use the Probability Mass Function (PMF)

$$P(X = x_i) = p(x_i) \qquad \sum p(x_i) = 1$$

Continuous Probability Distribution

- The probability is described over a range of values
- We use the Probability Density Function (PDF)

$$P(a \le X \le b) = \int_a^b f(x)dx \qquad \int_{-\infty}^{+\infty} f(x)dx = 1$$